

```
$ maxima -q
(%i1) load(finance)$ /* $ to suppress its output */
(%i2) fv(0.15, 10000, 10);
(%o2) 40455.57735707907
```

If you are interested in how exactly this is calculated, or what the formula is, Maxima is fun to play around with. Just use some symbols, instead of actual numbers:

```
(%i3) string(fv(r, p, n));
(%o3) p*(r+1)^n
(%i4) quit();
```

And, there you go. $p*(r+1)^n$ is the future value when investing 'p' amount for 'n' periods at the compounded interest rate of 'r' per period.

How about doing an inverse computation? Let's suppose, I want to get ₹1,000,000 after five years from my investment today at the interest rate of 10.75 per cent. Now, for that, how much should I invest today? Don't scratch your head, just call `pv(<rate>, <future_val>, <period>);`

```
(%i1) load(finance)$
(%i2) pv(0.1075, 1000000, 5);
(%o2) 600179.7323625274
(%i3) fv(0.1075, 600180, 5);
(%o3) 1000000.445928875
(%i4) quit();
```

So you need to invest ₹600,180 to get a value of ₹1,000,000 after five years at a rate of interest of 10.75 per cent – that's the check done above, using `fv()`.

Loans and EMIs

Today, with access to easy credit, people plan on buying a house, a car, or even smaller items, with the thought, "Let's buy it on loan and pay it off in EMIs (equated monthly instalments)." These terms would be familiar to most of you but how do you compute them? One assumes that there are some complicated formulae to do so, which is right. But, you don't need to worry about any of them. Just instruct Maxima to give you the complete schedule for your loan, at a given rate of interest, for a given period of time, using `amortization()`. The first example below provides the schedule for a home loan of ₹2 million at an interest rate of 9.25 per cent per annum (p.a.) for five years. The various columns in the output schedule provide the following information:

- n: instalment payment time – year, in the case of our example
- Balance: principal + interest left over to be paid out, after the current instalment is paid out
- Interest: interest part being paid out in the current instalment
- Amortization: principal part being paid out in the current instalment
- Payment: current instalment to be paid out, i.e., the EYI

(equated yearly instalment)

What is this EYI? We were supposed to talk only about EMI, right? Okay, in that case, we need to convert the rate of interest and the period, in terms of months. So, we need to divide the per annum rate of interest by 12 to get what it amounts to per month, and multiply the number of years by 12 to get that in the number of months. That is exactly what the second example below shows. Note that it is 60 EMIs to be paid out over the period of five years.

```
$ maxima -q
(%i1) load(finance)$
(%i2) amortization(0.0925/12, 2000000, 5)$
"n" "Balance" "Interest" "Amortization" "Payment"
0.000 2000000.000 0.000 0.000 0.000
1.000 1667475.420 185000.000 332524.580 517524.580
2.000 1304192.317 154241.476 363283.103 517524.580
3.000 907305.527 120637.789 396806.790 517524.580
4.000 473706.709 83925.761 433598.818 517524.580
5.000 0.000 43817.871 473706.709 517524.580
```

```
(%i3) amortization(0.0925/12, 2000000, 5*12)$
"n" "Balance" "Interest" "Amortization" "Payment"
0.000 2000000.000 0.000 0.000 0.000
1.000 1973656.870 15416.667 26343.130 41759.797
2.000 1947116.679 15213.605 26546.192 41759.797
3.000 1920359.860 15008.978 26750.818 41759.797
4.000 1893402.837 14802.774 26957.023 41759.797
5.000 1866238.021 14594.980 27164.816 41759.797
6.000 1838863.809 14385.585 27374.212 41759.797
7.000 1811278.588 14174.575 27585.221 41759.797
8.000 1783480.730 13961.939 27797.857 41759.797
9.000 1755468.598 13747.664 28012.133 41759.797
10.000 1727240.538 13531.737 28228.059 41759.797
11.000 1698794.007 13314.146 28445.651 41759.797
12.000 1670129.968 13094.877 28664.919 41759.797
13.000 1641244.090 12873.919 28885.878 41759.797
14.000 1612135.550 12651.257 29108.540 41759.797
15.000 1582892.632 12426.878 29332.918 41759.797
16.000 1553243.605 12200.770 29559.026 41759.797
17.000 1523456.728 11972.919 29786.877 41759.797
18.000 1493440.244 11743.312 30016.484 41759.797
19.000 1463192.382 11511.935 30247.861 41759.797
20.000 1432711.360 11278.775 30481.022 41759.797
21.000 1401995.381 11043.817 30715.980 41759.797
22.000 1371042.632 10807.048 30952.749 41759.797
23.000 1339851.289 10568.454 31191.343 41759.797
24.000 1308419.513 10328.020 31431.776 41759.797
25.000 1276745.450 10085.734 31674.063 41759.797
26.000 1244827.233 9841.580 31918.217 41759.797
27.000 1212662.979 9595.543 32164.253 41759.797
28.000 1180250.793 9347.610 32412.186 41759.797
29.000 1147588.763 9097.767 32662.030 41759.797
30.000 1114674.963 8845.997 32913.800 41759.797
```